

Pressure-Based Menu Selection on a Spherical Tangible Device

Sunbum Kim

ksb4587@kaist.ac.kr

HCI Lab, School of Computing, KAIST

Daejeon, Republic of Korea

YoungIn Kim

youngin@kaist.ac.kr

HCI Lab, School of Computing, KAIST

Daejeon, Republic of Korea

Geehyuk Lee

geehyuk@gmail.com

HCI Lab, School of Computing, KAIST

Daejeon, Republic of Korea

ABSTRACT

Spherical tangible devices are increasingly utilized in augmented and virtual reality applications due to their universal shape and ease of rotation. However, their form factor presents challenges in integrating input channels, such as buttons, leading to difficulties in menu selection and mode switching—functions essential for complex tasks. In this paper, we explored the utilization of air pressure to augment the input capabilities of spherical devices. We implemented linear menu selection using a pressure-sensitive spherical tangible device and evaluated its performance through three menu selection techniques: Dwell, Quick Release, and Stroke. Our results indicated that participants achieved an average success rate of 97.6% and an average selection time of 1,449 ms across all selection techniques for the 12 menu items. Dwell showed higher accuracy, while Quick Release and Stroke showed faster selection times. Lastly, we demonstrated the application of performing various tasks such as object manipulation, locomotion, and text input in a virtual reality environment using a spherical tangible device through pressure-based menu selection.

CCS CONCEPTS

• **Human-centered computing** → **Interaction techniques**; **Virtual reality**.

KEYWORDS

Force input, Pressure input, Menu selection, Tangible interaction, Spherical device

ACM Reference Format:

Sunbum Kim, YoungIn Kim, and Geehyuk Lee. 2024. Pressure-Based Menu Selection on a Spherical Tangible Device. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3613905.3651090>

1 INTRODUCTION

As Augmented Reality (AR) and Virtual Reality (VR) technologies have advanced and grown in popularity, their application across a broad spectrum of activities has significantly expanded. In such environments, the use of tangible objects to enable more effective and immersive interactions is becoming increasingly common, allowing users to engage with them as they would with everyday objects.

Among these, spherical tangible objects are particularly favored for their universal shape and ease of manipulation, such as rolling. These objects act as physical representations of elements within the AR/VR environment, thus enhancing the immersive experience. The versatility of these spherical devices spans various applications, including 3D object manipulation [9, 13, 26], locomotion [10, 11], data visualization [20], and educational purposes [16].

However, spherical tangible devices present certain inherent limitations. Their geometric configuration makes integrating input modalities, such as buttons, a significant challenge. Attaching a button to a specific spot requires the device to be rotated to a particular orientation for activation. Moreover, the button could interfere with the device's rolling capability. These complications make essential tasks like menu selection and mode switching difficult. Furthermore, even simple operations, such as selecting an object, become challenging. In response, prior research has explored the use of auxiliary hardware, including keyboards [1] and pedals [13], to provide additional inputs. For standalone use of the spherical device, solutions like incorporating a ring-shaped button [18], which is operable from any orientation, or utilizing tapping-based interactions [14], have been investigated. However, these solutions primarily address basic inputs, like the simple act of selecting an object.

We explored the integration of force input to enhance the input capabilities for a more practical and versatile application of spherical tangible devices. Force input has been extensively studied in the field of Human-Computer Interaction (HCI), particularly since the introduction of force-sensing technology in mobile touchscreen devices. It facilitates continuous input and rapid command execution through linear menus [4, 5, 12, 19, 24]. Given its ability to enable various inputs from a single point, force input seems particularly promising for spherical tangible devices with a single surface.

In this study, we aim to explore the potential of force input in enhancing the input capabilities of spherical tangible devices. We implemented linear menu selection using a pressure-sensitive spherical tangible device and evaluated its performance using three menu selection techniques: Dwell, Quick Release, and Stroke. Furthermore, we presented example scenarios that demonstrate mode switching through linear menu selection, thereby enabling the use of spherical tangible devices in various tasks, such as 3D object manipulation, locomotion, and text input.

2 RELATED WORK

2.1 Spherical Tangible Device

Tangible User Interfaces (TUIs) facilitate interaction through real-world objects. Within AR/VR environments, TUIs serve as tangible

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3651090>

proxies or representations of virtual objects or environments, enhancing immersive interactions across various applications. Spherical tangible devices have emerged as the most common form among the diverse TUIs. Due to their universal shape and ease of rolling, they are being utilized in various applications.

The most common application is 3D object manipulation, where mapping the movements of a spherical tangible device isomorphically to the movements of virtual objects allows for intuitive and effective manipulation. Early examples of this research can be seen in the works of Zhai et al. [26] and Hinckley et al. [13], who demonstrated the effectiveness of small, wired spherical input devices in docking tasks. Englmeier [9] presented a dwell-based interaction with a spherical proxy for performing 7-DOF (Degrees of Freedom) docking tasks, which include translation, rotation, and scaling. Another significant application is locomotion. Englmeier proposed techniques for both continuous [10] and discrete locomotion [11] using a spherical device. Satriadi et al. [21] utilized tangible globes for data visualization, showcasing use cases such as augmenting a tangible globe with virtual overlays and linking an augmented globe to an abstract data visualization.

Despite the diverse applications of spherical tangible devices, facilitating mode switching for these applications is not straightforward. Given the limited input modalities of spherical devices, their practical use needs careful consideration of mode-switching methods. Louis et al. [15] introduced interactions for using Hand-held Perspective-Corrected Displays (HPCDs) as general-purpose devices, enabling menu and command selection through widgets or radial menus.

In this study, we aim to explore new possibilities afforded by utilizing pressure modality on a spherical device, investigating how it can enhance its functionality and expand its application potential.

2.2 Pressure-Based Menu Selection

Early studies on pressure input utilized devices such as pens [17] and mice [4]. With the advent of pressure-sensitive mobile devices, numerous studies have been conducted in touchscreen environments. Force input has been applied in various applications, including the adjustment of slider values [5] and accessing out-of-reach areas [6]. However, one of the most prevalent applications is menu selection in linear menus [4, 5, 12, 19, 24]. These tasks are typically performed by manipulating a cursor that moves in response to force, thereby positioning it on the desired target and selecting it.

A focal area in menu selection research involves exploring various selection techniques. Among these, Dwell and Quick Release are regarded as particularly representative. Previous research indicated that while Dwell provided high accuracy, it resulted in longer target selection times. On the other hand, Quick Release enabled quicker target selection at the cost of reduced accuracy when compared to Dwell [19]. Corsten et al. [7] proposed an approach based on the Model Human Processor [3] to enhance the algorithm for recognizing target selection in the Quick Release technique. This enhancement resulted in fast target selection while maintaining accuracy comparable to that of the Dwell technique.

Another key area of interest in menu selection research is the mapping between visual cursor movement and force input. Although many studies have adopted a linear mapping approach,

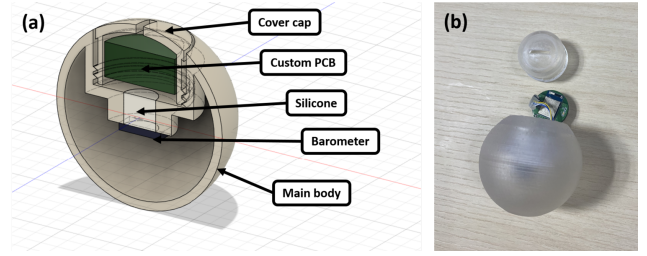


Figure 1: (a) The cross-sectional view of our spherical tangible device. (b) The main body, cover, and custom PCB.

there have been instances of employing quadratic mapping [4] or fish-eye functions [22]. Stewart et al. [23] emphasized the significance of understanding the sensor's characteristics and making linear adjustments before deciding on a specific mapping function.

While many studies have explored force input, applying their findings directly can be challenging due to differences in device form factors and sensing methods in our context. In this study, we aim to address this challenge by implementing a pressure-sensitive spherical tangible device and evaluating its menu selection capabilities, in line with previous research efforts.

3 PRESSURE-BASED MENU SELECTION ON A SPHERICAL TANGIBLE DEVICE

In this section, we describe the configuration of our pressure-sensitive spherical device and explain the pressure-based menu selection using this device.

3.1 Pressure-Sensitive Spherical Device

Figure 1 displays our pressure-sensitive spherical device, which consists of a main body and a detachable cover cap, with the entire sphere measuring 5.5 cm in diameter. The main body, designed with a hollow space, was 3D printed using flexible resin, while the cover was 3D printed with rigid resin. Within the main body, a barometer (BMP180) was placed. Below the cover cap, there existed a custom-printed circuit board (PCB) that incorporates a BLE module (nRF52840, Nordic) and a 9-axis motion sensor (ICM20948, InvenSense). The barometer was connected to the PCB via wired connections, and pressure values were transmitted using I2C communication. The BLE module then sent these pressure values to a PC through Bluetooth. The entrance to the main body was sealed with silicone to ensure the device remained air-tight.

3.2 Pressure-Based Menu Selection Techniques

Menu selection using pressure input typically involves manipulating a cursor's position through applied force and confirming the intended target. Among the various methods for confirmation, Dwell and Quick Release are the most prevalent. The Dwell method requires keeping the cursor on a target for a specific duration to confirm the selection, while Quick Release involves a rapid decrease in pressure after positioning the cursor on the target. Findings from comparisons between these two methods have varied across studies, likely due to differences in their implementation or the

specific contexts of each study. Additionally, in this study, we introduced a modified version of the Stroke [19], which utilizes a quick wrist rotation for confirmation. Employing a wrist rotation gesture for Stroke, rather than a translational gesture, may offer a more effective solution for the spherical tangible device.

Building on the findings of Corsten et al. [7], who applied the CMN model to the Quick Release method with an offset time of 240 ms to enhance accuracy, our study further investigates the impact of applying a 240 ms offset to both Quick Release and Stroke methods. For Quick Release, target selection is confirmed based on the cursor's position 240 ms before returning to the default pressure. For Stroke, selection is confirmed based on the cursor's position 240 ms before exceeding a specified rotation speed. The Dwell method was implemented with a dwell time of 1 second.

The performance of menu selection can be influenced by both the force range used and the visual mapping method. Based on our pilot study, we established a maximum force range of 200 hPa above the default state. This range required a considerable amount of force to select the topmost menu item. Applying pressure exceeding 200 hPa allows the cursor to move beyond the menu range, making the selection of the top target with proper force. The visual cursor was linearly mapped to the pressure values. The default sensor value showed a variation of approximately 10 hPa around 1013 hPa. To mitigate unintended forces that might occur while holding the device, we excluded the lower 20 hPa force range from the menu.

4 USER STUDY

This study aims to assess the performance of three menu selection techniques using our pressure-sensitive spherical tangible device.

4.1 Participants

We recruited 12 participants ($M = 22.25$, $SD = 2.9$, 1 female and 11 males) from our university's online community. One participant was left-handed, and the remainder were right-handed.

4.2 Experimental Setup

The experimental program was implemented in the Unity 2019.4.23f environment. The task was displayed on a 24-inch monitor, and participants performed the task while seated approximately one meter away. Figure 2(a) shows the experimental setup.

4.3 Design

We used a within-subject design with three independent variables: Technique (*Dwell*, *Quick Release*, *Stroke*), Menu Count—the number of targets (4, 8, 12), and Force Level—the force levels required for selection (45, 90, 135, 180). This design follows the conventions of traditional force input studies [4, 19, 24]. Figure 2(b) illustrates the target positions for each Menu Count. We measured *selection time* and *success rate* as dependent variables. Additionally, for assessing subjective preference, akin to the approach by Fruchard et al. [12], we gathered data on subjective preference using a 7-point Likert scale, evaluating perceived *comfort* (1-very uncomfortable to 7-very comfortable), *accuracy* (1-very inaccurate to 7-very accurate), and *physical demand* (1-very demanding to 7-not demanding) for each technique.

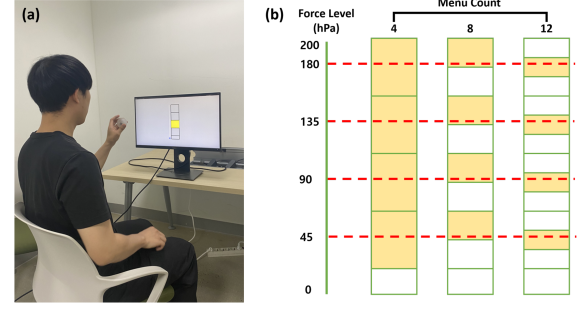


Figure 2: (a) Experimental Setup in User Study. (b) Representation of the interaction between the Menu Count and Force Level independent variables. The yellow cells are the targets used in the experiment.

4.4 Procedure

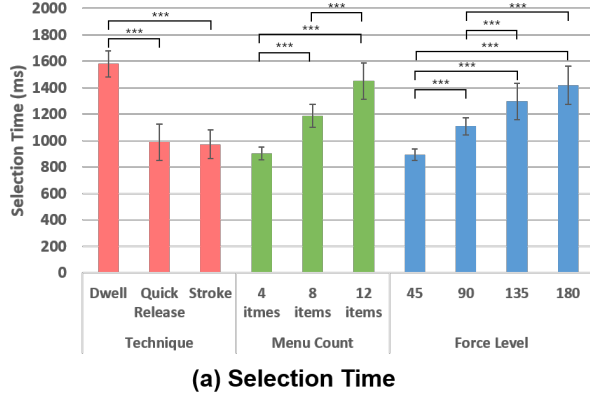
The participant was guided into a room and seated at a desk. The experimenter explained the first technique condition. Then, the participant engaged in a 2-minute practice session. After this session, the participant performed the main task, which involved selecting targets as quickly and accurately as possible. Upon completing the task under the first condition, the participant took a 1-minute break. This process was then repeated for the remaining two conditions. Following all conditions, participants filled out a questionnaire survey, and the experimenter conducted brief interviews. The order of the Technique was completely counterbalanced among participants. Menu Count and Force Level were randomized within each technique condition. In total, we collected 2,160 data points (12 participants $\times$ 3 Techniques $\times$ 3 Menu Counts $\times$ 4 Force Levels $\times$ 5 trials). Each experiment lasted approximately 45 minutes.

4.5 Results

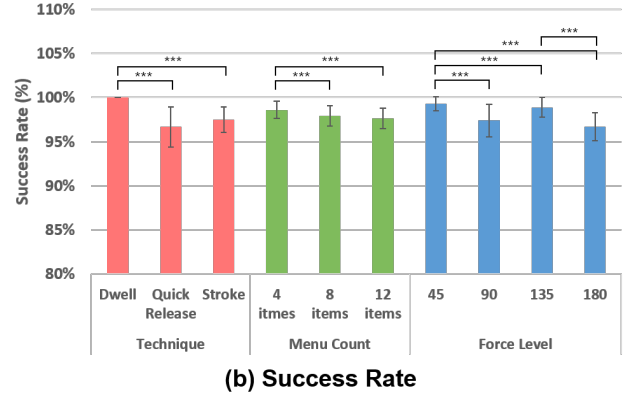
To examine the statistical significance of *selection time* and *success rate*, we conducted a three-way repeated measures ANOVA. Since *selection time* and *success rate* did not meet the normality assumption, we performed an Aligned Rank Transform (ART) [25] prior to the ANOVA. For the post-hoc analysis, we employed the ART-C procedure as proposed by Elkin et al. [8]. We conducted a Friedman test and post-hoc Wilcoxon Signed-Rank tests to analyze subjective preference data. The post-hoc tests were Bonferroni corrected, and the post-hoc results are reported with corrected p-values.

4.5.1 Selection Time. Figure 3(a) presents an overview of the results for *selection time*. An ANOVA test revealed significant main effects for Technique ($F_{(2,22)} = 38.172$, $p < 0.001$), Menu Count ($F_{(2,22)} = 235.802$, $p < 0.001$), and Force Level ($F_{(3,33)} = 82.808$, $p < 0.001$). Significant interactions were observed for Technique $\times$ Menu Count ($F_{(4,44)} = 4.431$, $p < 0.001$), Technique $\times$ Force Level ($F_{(6,66)} = 2.974$, $p < 0.001$), Menu Count $\times$ Force Level ($F_{(6,66)} = 13.509$, $p = 0.012$), and the three-way Technique $\times$ Menu Count $\times$ Force Level interaction ($F_{(12,132)} = 2.296$, $p = 0.011$).

The purpose of this study was to explore the performance of various selection techniques using our prototype and to determine the most effective technique. Therefore, post-hoc results were reported briefly. Post-hoc tests indicated that *Dwell* was significantly



(a) Selection Time



(b) Success Rate

Figure 3: Results in User Study: (a) selection time and (b) success rate. Error bars indicate 95% confidence intervals, and asterisks indicate significant differences (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$).**

slower than both *Quick Release* ($p < 0.001$) and *Stroke* ($p < 0.001$). Significant differences were found between all Menu Count conditions ($p < 0.001$), as well as between all Force Level conditions ($p < 0.001$), except between 135 and 180 ($p = 0.055$). As the number of menu items increased and the force level became higher, selection time increased.

4.5.2 Success Rate. Figure 3(b) presents an overview of results for success rate. An ANOVA test revealed significant main effects for Technique ($F_{(2,22)} = 35.579$, $p < 0.001$), Menu Count ($F_{(2,22)} = 32.779$, $p < 0.001$), and Force Level ($F_{(3,33)} = 29.358$, $p < 0.001$). Significant interactions were observed for Technique $\times$ Menu Count ($F_{(4,44)} = 15.741$, $p < 0.001$), Technique $\times$ Force Level ($F_{(4,44)} = 12.703$, $p < 0.001$), Menu Count $\times$ Force Level ($F_{(6,66)} = 10.566$, $p < 0.001$), and the three-way Technique $\times$ Menu Count $\times$ Force Level interaction ($F_{(12,132)} = 5.974$, $p < 0.001$).

Post-hoc test results indicated that *Dwell* had a significantly higher success rate than both *Quick Release* ($p < 0.001$) and *Stroke* ($p < 0.001$). Significant differences were found between 4-item Menu Size and both 8-item ($p < 0.001$) and 12-item ($p < 0.001$) Menu Count conditions. The post-hoc results for Force Level are presented in Figure 3(b).

4.5.3 Questionnaire & Preference. Table 1 presents the results of the questionnaire survey. A Friedman test revealed that Technique significantly affected *comfort* ($\chi^2(2) = 10.757$, $p = 0.005$), *accuracy* ($\chi^2(2) = 17.591$, $p < 0.001$), and *physical demand* ($\chi^2(2) = 11.209$, $p = 0.004$). The post-hoc analysis showed that *Quick Release* was perceived as the most comfortable (vs. *Dwell*, $Z = -2.567$, $p = 0.031$; vs. *Stroke*, $Z = -2.598$, $p = 0.028$), and *Dwell* was perceived as the most accurate (vs. *Quick Release*, $Z = -2.956$, $p = 0.009$; vs. *Stroke*, $Z = -2.970$, $p = 0.009$). However, no significant differences were found in *physical demand* between any of the pairs. Table 2 presents preference rank results for each Technique. A Friedman test indicated no significant difference ($\chi^2(2) = 0.500$, $p = 0.779$).

4.6 Summary & Discussion

The results from our study using the pressure-sensitive spherical device are consistent with findings from previous studies. The

Question	Technique			Friedman Test		
	Dwell	Quick Release	Stroke	$\chi^2(2)$	p -value	Post-hoc
Comfort	4.42 (1.04)	6.33 (0.37)	5.42 (0.66)	10.757	0.005	D-QR, QR-S
Accuracy	6.75 (0.35)	4.58 (0.66)	5.67 (0.61)	17.591	<0.001	D-QR, D-S
Physical Demand	3.08 (0.74)	5.08 (0.78)	4.08 (0.61)	11.209	0.004	no pair

Table 1: Mean subjective rating scores in User Study. The values inside parentheses indicate 95% confidence intervals.

	First	Second	Third
Dwell	4	3	5
Quick Release	5	1	6
Stroke	3	8	1

Table 2: Results of a preference survey. 'First' indicates the most preferred condition, while 'Third' indicates the least preferred.

Dwell technique, despite requiring more time for selection than *Quick Release* and *Stroke*, achieved a 100% success rate. We observed that an increase in Menu Count resulted in longer selection times and reduced success rates. The overall performance of each technique was comparable to or better than those reported in previous studies.

Based on our preliminary investigation and previous studies, we opted for menu counts of 4, 8, and 12. However, the high accuracy achieved by participants, even with 12-menu items, indicates the feasibility of exploring larger menu counts in future studies to better understand menu selection performance.

Although significant differences in selection time and success rate were observed across the techniques, these differences were relatively modest. The *Dwell* technique required approximately 0.6 seconds more than the others, but its 100% success rate makes it suitable for tasks such as command input and menu selection. Moreover, reducing the dwell time from the 1 second used in this

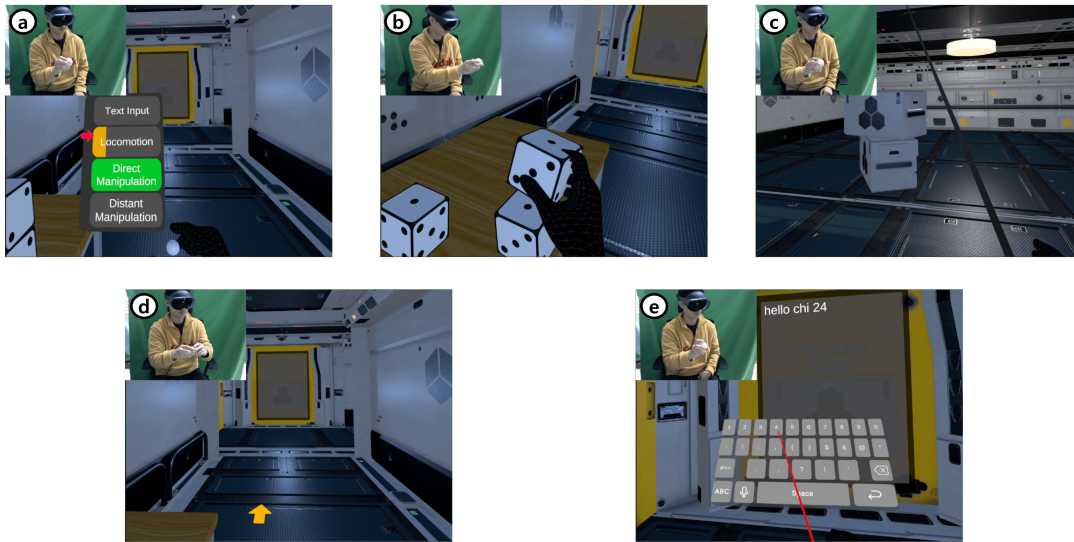


Figure 4: Our example scenario demonstrates various functionalities: (a) mode switching using pressure-based menu selection, (b) direct object manipulation, (c) distant object manipulation, (d) locomotion, and (e) text input

study to the 0.5 seconds used in other studies could further reduce selection time.

Participants' preferences for the techniques varied based on individual criteria. Those who preferred *Dwell* appreciated its clear visual feedback, though some noted the challenge of maintaining a grip for one second. *Quick Release* was favored for its elimination of the need for sustained force and the avoidance of additional gestures, unlike *Stroke*. However, some participants highlighted uncertainties about target selection as a disadvantage. *Stroke* was perceived as an intermediate technique between the other two techniques in terms of confidence in target selection and fatigue.

5 EXAMPLE SCENARIO

To present the versatile use of spherical tangible devices through pressure-based menu selection, we implemented example scenarios demonstrating fundamental functionalities: direct object manipulation, distant object manipulation, locomotion, and text input. Figure 4 shows our example scenario scene. Users can invoke the menu by pressing the spherical tangible device and select a mode using the dwell technique. In direct object manipulation, users can grasp a virtual object by approaching it and performing a squeeze-and-release gesture. The movements of the virtual object are then directly mapped to those of the spherical tangible device in a one-to-one mapping. In distant object manipulation, users can interact with virtual objects from afar. By pointing at an object through raycasting and performing a squeeze-and-release gesture, users can grasp the object. This method of distant manipulation utilizes the direct HOMER technique [2]. Locomotion is achieved by rolling the device forward or backward to move in the respective direction, and by rolling it sideways to execute left or right turns. In text input mode, a virtual keyboard becomes accessible, allowing characters to be selected and entered through raycasting, followed by a squeeze-and-release gesture. Our demonstration application seamlessly supports these diverse tasks, facilitating smooth mode

switching through pressure-based menu selection without any conflicts.

6 CONCLUSION

In this study, we explored the performance of pressure-based menu selection using a spherical tangible device. In alignment with prior studies on pressure-based menu selection, our findings demonstrated that the *Dwell* technique provided enhanced accuracy, whereas the *Quick Release* and *Stroke* techniques offered advantages in terms of reduced selection times. Participants achieved a notable average success rate of 97.6% and an average selection time of 1,449 milliseconds across all techniques for the 12 menu items. Participant preferences for selection methods exhibited variation. Moreover, we developed a demonstration application that utilizes pressure-based menu selection to perform various functions on the spherical device.

This study serves as an initial investigation into the feasibility of enhancing a spherical tangible device with pressure input to function as a practical controller. Future research could investigate diverse interaction techniques that utilize pressure or examine strategies to improve the utility of the spherical tangible device through other input modalities.

ACKNOWLEDGMENTS

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT) (No.2022-0-00137, XR User Interaction Evaluation and Employing Its Technology)

REFERENCES

- [1] Lonni Besançon, Paul Issartel, Mehdi Ammi, and Tobias Isenberg. 2017. Mouse, Tactile, and Tangible Input for 3D Manipulation. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 4727–4740. <https://doi.org/10.1145/3025453.3025863>

- [2] Doug A. Bowman and Larry F. Hodges. 1997. An Evaluation of Techniques for Grabbing and Manipulating Remote Objects in Immersive Virtual Environments. In *Proceedings of the 1997 Symposium on Interactive 3D Graphics* (Providence, Rhode Island, USA) (I3D '97). Association for Computing Machinery, New York, NY, USA, 35–ff. <https://doi.org/10.1145/253284.253301>
- [3] Stuart K Card. 2018. *The psychology of human-computer interaction*. Crc Press.
- [4] Jared Cechanowicz, Pourang Irani, and Sriram Subramanian. 2007. Augmenting the Mouse with Pressure Sensitive Input. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '07). Association for Computing Machinery, New York, NY, USA, 1385–1394. <https://doi.org/10.1145/1240624.1240835>
- [5] Christian Corsten, Bjoern Daehlmann, Simon Voelker, and Jan Borchers. 2017. BackXPress: Using Back-of-Device Finger Pressure to Augment Touchscreen Input on Smartphones. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 4654–4666. <https://doi.org/10.1145/3025453.3025565>
- [6] Christian Corsten, Marcel Lahaye, Jan Borchers, and Simon Voelker. 2019. ForceRay: Extending Thumb Reach via Force Input Stabilizes Device Grip for Mobile Touch Input. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3290605.3300442>
- [7] Christian Corsten, Simon Voelker, and Jan Borchers. 2017. Release, Don't Wait! Reliable Force Input Confirmation with Quick Release. In *Proceedings of the 2017 ACM International Conference on Interactive Surfaces and Spaces* (Brighton, United Kingdom) (ISS '17). Association for Computing Machinery, New York, NY, USA, 246–251. <https://doi.org/10.1145/3132272.3134116>
- [8] Lisa A. Elkin, Matthew Kay, James J. Higgins, and Jacob O. Wobbrock. 2021. An Aligned Rank Transform Procedure for Multifactor Contrast Tests. In *The 34th Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '21). Association for Computing Machinery, New York, NY, USA, 754–768. <https://doi.org/10.1145/3472749.3474784>
- [9] David Englmeier, Julia Dörner, Andreas Butz, and Tobias Höllerer. 2020. A Tangible Spherical Proxy for Object Manipulation in Augmented Reality. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, 221–229. <https://doi.org/10.1109/VR46266.2020.00041>
- [10] David Englmeier, Fan Fan, and Andreas Butz. 2020. Rock or Roll – Locomotion Techniques with a Handheld Spherical Device in Virtual Reality. In *2020 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 618–626. <https://doi.org/10.1109/ISMAR50242.2020.00089>
- [11] David Englmeier, Wanja Sajko, and Andreas Butz. 2021. Spherical World in Miniature: Exploring the Tiny Planets Metaphor for Discrete Locomotion in Virtual Reality. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*. IEEE, 345–352. <https://doi.org/10.1109/VR50410.2021.00057>
- [12] Bruno Fruchard, Paul Strohmeier, Roland Bennewitz, and Jürgen Steimle. 2021. Squish This: Force Input on Soft Surfaces for Visual Targeting Tasks. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 219, 9 pages. <https://doi.org/10.1145/3411764.3445623>
- [13] Ken Hinckley, Joe Tullio, Randy Pausch, Dennis Proffitt, and Neal Kassell. 1997. Usability Analysis of 3D Rotation Techniques. In *Proceedings of the 10th Annual ACM Symposium on User Interface Software and Technology* (Banff, Alberta, Canada) (UIST '97). Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/263407.263408>
- [14] Sunbum Kim, Youngbo Aram Shim, and Geehyuk Lee. 2022. Exploration of Form Factor and Bimanual 3D Manipulation Performance of Rollable In-Hand VR Controller. In *Proceedings of the 28th ACM Symposium on Virtual Reality Software and Technology* (Tsukuba, Japan) (VRST '22). Association for Computing Machinery, New York, NY, USA, Article 15, 11 pages. <https://doi.org/10.1145/3562939.3565625>
- [15] Thibault Louis, Jocelyne Troccaz, Amélie Rochet-Capellan, and François Bérard. 2020. GyroSuite: General-Purpose Interactions for Handheld Perspective Corrected Displays. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '20). Association for Computing Machinery, New York, NY, USA, 1248–1260. <https://doi.org/10.1145/3379337.3415893>
- [16] Hovhannes Mayilyan. 2019. Augmented reality in education, ar globe project assessment in actual teaching-learning environment. *International Journal of Learning, Teaching and Educational Research* 18, 3 (2019), 1–14.
- [17] Sachi Mizobuchi, Shinya Terasaki, Turo Keski-Jaskari, Jari Nousiainen, Matti Ryyanen, and Miika Silfverberg. 2005. Making an Impression: Force-Controlled Pen Input for Handheld Devices. In *CHI '05 Extended Abstracts on Human Factors in Computing Systems* (Portland, OR, USA) (CHI EA '05). Association for Computing Machinery, New York, NY, USA, 1661–1664. <https://doi.org/10.1145/1056808.1056991>
- [18] Gary Perelman, Marcos Serrano, Mathieu Raynal, Celia Picard, Mustapha Derras, and Emmanuel Dubois. 2015. The Roly-Poly Mouse: Designing a Rolling Input Device Unifying 2D and 3D Interaction. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) (CHI '15). Association for Computing Machinery, New York, NY, USA, 327–336. <https://doi.org/10.1145/2702123.2702244>
- [19] Gonzalo Ramos, Matthew Boulos, and Ravin Balakrishnan. 2004. Pressure Widgets. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vienna, Austria) (CHI '04). Association for Computing Machinery, New York, NY, USA, 487–494. <https://doi.org/10.1145/985692.985754>
- [20] Kadek Ananta Satriadi, Barrett Ens, Tobias Czauderna, Maxime Cordeil, and Bernhard Jenny. 2021. Quantitative Data Visualisation on Virtual Globes. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 460, 14 pages. <https://doi.org/10.1145/3411764.3445152>
- [21] Kadek Ananta Satriadi, Jim Smiley, Barrett Ens, Maxime Cordeil, Tobias Czauderna, Benjamin Lee, Ying Yang, Tim Dwyer, and Bernhard Jenny. 2022. Tangible Globes for Data Visualisation in Augmented Reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 505, 16 pages. <https://doi.org/10.1145/3491102.3517715>
- [22] Kang Shi, Pourang Irani, Sean Gustafson, and Sriram Subramanian. 2008. PressureFish: A Method to Improve Control of Discrete Pressure-Based Input. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Florence, Italy) (CHI '08). Association for Computing Machinery, New York, NY, USA, 1295–1298. <https://doi.org/10.1145/1357054.1357256>
- [23] Craig Stewart, Michael Rohs, Sven Kratz, and Georg Essl. 2010. Characteristics of Pressure-Based Input for Mobile Devices. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Atlanta, Georgia, USA) (CHI '10). Association for Computing Machinery, New York, NY, USA, 801–810. <https://doi.org/10.1145/1753326.1753444>
- [24] Graham Wilson, Craig Stewart, and Stephen A. Brewster. 2010. Pressure-Based Menu Selection for Mobile Devices. In *Proceedings of the 12th International Conference on Human Computer Interaction with Mobile Devices and Services* (Lisbon, Portugal) (MobileHCI '10). Association for Computing Machinery, New York, NY, USA, 181–190. <https://doi.org/10.1145/1851600.1851631>
- [25] Jacob O. Wobbrock, Leah Findlater, Darren Gergle, and James J. Higgins. 2011. The Aligned Rank Transform for Nonparametric Factorial Analyses Using Only Anova Procedures. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) (CHI '11). Association for Computing Machinery, New York, NY, USA, 143–146. <https://doi.org/10.1145/1978942.1978963>
- [26] Shumin Zhai, Paul Milgram, and William Buxton. 1996. The Influence of Muscle Groups on Performance of Multiple Degree-of-Freedom Input. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, British Columbia, Canada) (CHI '96). Association for Computing Machinery, New York, NY, USA, 308–315. <https://doi.org/10.1145/238386.238534>